

Todd Takala

Lead Data Scientist, Machine Learning Engineer, AI/ML and Analytics

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Summary

Data Scientist & Machine Learning Engineer with strong leadership in machine learning and statistical modeling who generated over \$1.3 billion in business value. Specializations include design of data solutions, implementation of MLOps pipelines on AWS and Snowflake, enhancing customer engagement.

Professional Experience

Lead Data Scientist — Caterpillar Inc. — Surprise, AZ — 09/2022 to Present

- Architected and developed end-to-end data products using programming skills in Python and R, including an award-winning AI platform to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award
- Collaborated with senior leadership and stakeholders to facilitate understanding of the project's broad impact and create precise user requirements, ensuring alignment with the business road map through written and verbal communication
- Performed data mining on structured, semi-structured, and unstructured data to assemble a training dataset from multiple sources, considering thousands of parameters derived from dealer and product information
- Used LLMs and natural language processing to find similarities in service jobs and performed feature engineering to retain important parameters, which increased model accuracy by 40%
- Conducted exploratory data analysis on the training datasets, eliminating outliers using sophisticated analytics such as isolation forest and anomaly detection techniques
- Trained AI models using Python and SageMaker, applying machine learning algorithms like neural networks, PyTorch, XGBoost, TensorFlow, and LightGBM, which improved model performance and enhanced the accuracy of equipment repair time predictions, leading to better maintenance scheduling
- Engineered production-ready MLOps pipelines on AWS and Snowflake, establishing best practices for cloud architecture. Managed project lifecycles, automated enterprise networks, and implemented monitoring strategies to assess model performance and business outcomes.
- Implemented robust information management strategies, such as STAR schema, to ensure data integrity and accessibility for the Snowflake data lake, improving data reliability and supporting informed decision-making

Reliability Analytics Manager — Amazon.com — Tempe, AZ — 02/2022 to 09/2022

- Applied advanced analytics, mathematics, and statistics to create a new KPI for autonomous robots used in warehouse automation through big data analysis, and wrote a white paper on statistical programming, resulting in its adoption by senior management
- Led a team of business intelligence engineers and data analysts to deploy a pipeline using Shiny R, which improved real-time communication of robot status at all global fulfillment centers, enhancing operational efficiency and data analytics capabilities
- Utilized advanced data science techniques to extract actionable insights from research to driving strategic decision-making and improving operational outcomes
- Designed and implemented solutions utilizing large databases with Redshift and PostgreSQL, supporting data-driven decision-making and enhancing business intelligence capabilities

Data Engineering Manager — Empire-Cat — Mesa, AZ — 01/2012 to 02/2022

- Created predictive ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to \$62 million in savings within 90 days
- Developed a time-series MLOps pipeline for AI-driven forecasting in supply chain management, integrated data using SQL and Pandas, managed a component exchange program, and collaborated with clients on durability forecasting, enhancing supply chain efficiency
- Supervised 8 engineers and acted as a customer-facing expert on mining equipment. Implemented a data-driven failure analysis program, boosting an aging fleet's physical availability to over 90%.
- Applied the DMAIC Six Sigma methodology and computer science with complex problem-solving to identify opportunities for improving operation, manufacturing, and maintenance of equipment fleet, yielding \$1.2 billion in savings over 6 years as Reliability Engineering Manager prior to promotion
- Leadership experience with business acumen to build trust and enhance customer experience and relationships to promote data-based decisions. Partnered on design of experiments to improve reliability, safety and durability

Certification & Technical Registration

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **BerkeleyX:** The Science of Happiness at Work Professional Certificate
- **Canonical:** Ubuntu Linux Professional Certificate
- **Data Camp:** AI Fundamentals Certificate, Certified SQL Associate, Data Scientist Professional Certificate
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IBM:** Advanced Deep Learning Specialist, RAG and Agentic AI Professional Certificate
- **Komatsu North America:** Certified Technical Communicator
- **US Department of Labor:** Operating Engineer Vocational Certificate

Education

Bachelor of Science in Mechanical Engineering — Arizona State University — Tempe, AZ — 06/2003 to 05/2006
STEM, Multivariate Calculus, Differential Equations, Linear Algebra, Numerical Methods, Algorithms, Data Structures, Statistics, Mechanics, Material Science, Manufacturing, Finite Element Analysis, Computational Fluid Dynamics, Internal Combustion Engines, Mechanisms, Heat Transfer, Thermodynamics, Manufacturing, Economics, Cost, Finance, Control Systems, Design

Hard Skills

A/B Testing, Agentic AI, Agile Methodology, AI/ML, Anomaly Detection, Amazon Bedrock, Artificial Intelligence, AWS, BASH, CI/CD, Computer Science, RDBMS, Database Administrator, Data Science, DBT, Deep Learning, Design of Experiments, DevOps, Docker, Embedding, ETL, Fine Tuning, Advanced Excel, Financial Forecasting, Generative AI, Git, Linear Algebra, Large Language Models, LLM, Mathematics, Natural Language Processing, NLP, Pandas, Power BI, Public Cloud, Python, PyTorch, R, RAG, Research, Sagemaker, Scikit-learn, Scripting Language, SDLC, Simulation, SnapLogic, Snowflake, SQL Queries, Statistical Analysis, Streamlit, SQL, Tableau, TensorFlow, Transact SQL, Transformers, Version Control

Soft Skills

Adaptability, Attention to Detail, Business Requirements, Coaching, Communication Skills, Continuous Improvement, Creative Problem Solving, Data Driven, Decision making, Growth Mindset, Interviewing, Leadership Experience, Life Long Learning, Listening, Mentoring, Methodical, Multicultural Collaboration, Open minded, Problem Solving, Professional Services, Presentations, Public Speaking, Strategic Thinking, Team Building, Thought Leadership, Willing to Learn